

AMD DEVELOPER HACKATHON

TRACK 3 UNICORN

Crossfire CUDA to ROCm Translation Agent

An autonomous AI agent that ports CUDA code to AMD ROCm — HIPIFY-first migration, compiled and test-verified on AMD MI300X.

AMD CLOUD

FIREWORKS AI

GEMMA 4 12B

ROCM 7.2

THE PROBLEM

AMD has the hardware.
NVIDIA has the **software moat.**
We **breach it.**

Multi-billion-dollar

AI INFRA MARKET LOCKED TO NVIDIA

Tens of thousands of ML teams want to migrate to AMD but cannot afford the rewrite.

2-8 wks

MANUAL CUDA-TO-ROCM PORT TIME

Per project. Every PyTorch extension, every custom kernel, every inference pipeline.

WHY NOW

Four forces converge in 2026

1. ROCm 7.2 is stable

ROCm has matured rapidly for production AI workloads. MIOpen, hipBLAS, rocPRIM are production-ready.

2. MI300X offers compelling inference economics

192GB HBM3, lower cost per token. The hardware advantage is real — Software portability remains a major barrier to AMD adoption.

3. Code LLMs are viable

Gemma 4, DeepSeek-Coder, and other modern code LLMs perform semantic code translation with RAG augmentation.

4. AMD is prioritizing software portability

Lisa Su: AMD continues to prioritize AI software and developer tooling. ROCm roadmap lists "CUDA porting tools" as priority.

THE SOLUTION

Autonomous translation + validation

Crossfire takes supported CUDA source files, translates them to HIP, compiles and executes them on AMD MI300X, and test-verifies the outputs against analytical reference baselines — autonomously, in minutes rather than weeks.



ARCHITECTURE

AMD-native execution and validation stack

AMD

Agent Orchestrator

LangGraph state machine on AMD Developer Cloud. 6-node loop with retry.

AMD

Local Model

Gemma 4 12B via vLLM on MI300X. \$0 external API cost for local routes. LoRA fine-tuning planned.

AMD

Cloud Fallback

Gemma 4 12B via vLLM on MI300X for hard semantic cases (cuDNN, Triton, warp shuffle).

GEMMA

RAG Layer

ChromaDB with ROCm 7.2 docs, MIOpen API refs, translation tables.

AMD

Compile Sandbox

Docker + ROCm 7.2.3 + hipcc. Dockerized ROCm compile environment.

AMD

Validation

Each supported demo is compiled with hipcc, executed on AMD MI300X, and numerically compared against an analytical reference output.

LIVE DEMO

Watch it work

5

HERO DEMOS

From vector_add to warp-shuffle reduction

Each demo shows a distinct CUDA pattern: simple kernel, shared memory tiling, cuBLAS library call, PyTorch extension, warp shuffle primitives.

- 01

vector_add

Easy · 0.30
- 02

matrix_mul

Medium · 0.45
- 03

sgemm (cuBLAS)

Hard · 0.65
- 04

pytorch_ext

Hard · 0.70
- 05

warp_shuffle

Hardest · 0.90

→ Live demo at: 129.212.185.42:8000/ui

VALIDATION

Test-verified on **AMD MI300X**

Every translation is compiled with hipcc, run on AMD MI300X, and numerically diffed against analytical baseline outputs. Pass threshold: $\text{max_abs_error} < 1\text{e-}5$.

1e-5

DIFF THRESHOLD

5

MAX ITERATIONS

80%

LOCAL MODEL USAGE

\$0

EXTERNAL API COST

SAMPLE VALIDATION OUTPUT

```
$ crossfire translate vector_add.cu
✓ Analyzed: 1 kernel, difficulty=0.30, routing=LOCAL
✓ Translated: 1728 chars, 0 tokens (local, FREE)
✓ Compiled: hipcc success in 1.2s
✓ Ran on MI300X: 0.45ms
✓ Diff: max_abs_error=0.0e+00 → PASS
✓ Done in 2.1s, 1 iteration
```

TECHNICAL DIFFERENTIATION

HIPIFY-first + Gemma 4 12B semantic repair (LoRA pipeline on roadmap)

Approach	Syntax	Semantics	Automated test verification	Time	AMD-Native
Manual rewrite	✓	✓	✓	2-8 wks	Optional
AMD HIPIFY (rule-based)	✓	✗	✗	Hours	✓
GitHub Copilot (generic)	Partial	Partial	✗	Hours	✗
Crossfire v1 prototype	✓	✓	✓	Minutes	✓

→ Crossfire combines semantic AI translation with autonomous validation on AMD hardware.

MARKET POTENTIAL

Multi-billion-dollar market. A strategic bridge into the AMD ecosystem.

The Buyer

ML platform teams at mid-to-large enterprises evaluating AMD hardware and facing CUDA migration friction.

The Revenue Model

Free tier (5 files/mo), Pro (\$99/mo), Enterprise (custom). Plus AMD white-label / OEM licensing.

Strategic Fit

AMD has acquired Mipsology, Nod.ai, Silo AI — all software gap-closers. Crossfire fits the pattern exactly.

The Moat

Validated translation corpus grows with every job. Fine-tuned model improves. Each migration adds reusable repair traces and validated mappings.

EXECUTION

Built in 5 days. Imagine 5 months.

D1

Kickoff
First translation

D2

Agent core
RAG + routing

D3

Sandbox
ROCm compile + run

D4

Frontend
Demo + video

D5

Polish
Edge cases

D6

Submit
v1.0.0

Roadmap

v1.0 (Hackathon)

5 hero demos, single-file, prompt-engineered baseline

v2.0 (3 months)

Multi-file repos, fine-tuned model, 50K training pairs

v3.0 (12 months)

Triton support, cuDNN full coverage, enterprise SaaS

Crossfire is AMD's strategic unlock.

We built it in 5 days. Imagine what we build in 5 months with AMD.

[AMD CLOUD](#)[FIREWORKS](#)[GEMMA 4 12B](#)[ROCM 7.2](#)

github.com/VampFay/CROSSFIRE_AMD-DEV-HACKATHON-ACT_2 · 129.212.185.42:8000/ui